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Contactless WiFi Sensing and Monitoring for Future Healthcare: Emerging Trends, Challenges and Opportunities

Yao Ge, Ahmad Taha, [S. A. Shah](#), Kia Dashtipour, Shuyuan Zhu, Jonathan M. Cooper, Qammer Abbasi, Muhammad Imran

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Abstract

WiFi sensing has recently received significant interest from academics, industry, healthcare professionals and other caregivers (including family members) as a potential mechanism to monitor our aging population at distance, without deploying devices on users bodies. In particular, these

methods have gained significant interest to efficiently detect critical events such as falls, sleep disturbances, wandering behavior, respiratory disorders, and abnormal cardiac activity experienced by vulnerable people. The interest in such WiFi-based sensing systems stems from its practical deployments in indoor settings and compliance from monitored persons, unlike other sensors such as wearables, camera-based, and acoustic-based solutions. This paper reviews state-of-the-art research on collecting and analysing channel state information, extracted using ubiquitous WiFi signals, describing a range of healthcare applications and identifying a series of open research challenges, untapped areas, and related trends. This work aims to provide an overarching view in understanding the technology and discusses its uses-cases from a perspective that considers hardware, advanced signal processing, and data acquisition.

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Keywords

- Activity recognition
- Biomedical monitoring
- deep learning
- healthcare detection
- machine learning
- Market research
- Medical services
- Monitoring
- Sensors

- WiFi sensing
- Wireless fidelity

ASJC Scopus subject areas

- Biomedical Engineering

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keywords = "Activity recognition, Biomedical monitoring, deep learning, healthcare detection, machine learning, Market research, Medical services, Monitoring, Sensors, WiFi sensing, Wireless fidelity",

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